

Predictive Maintenance — Analytical Findings & Technical Interpretations

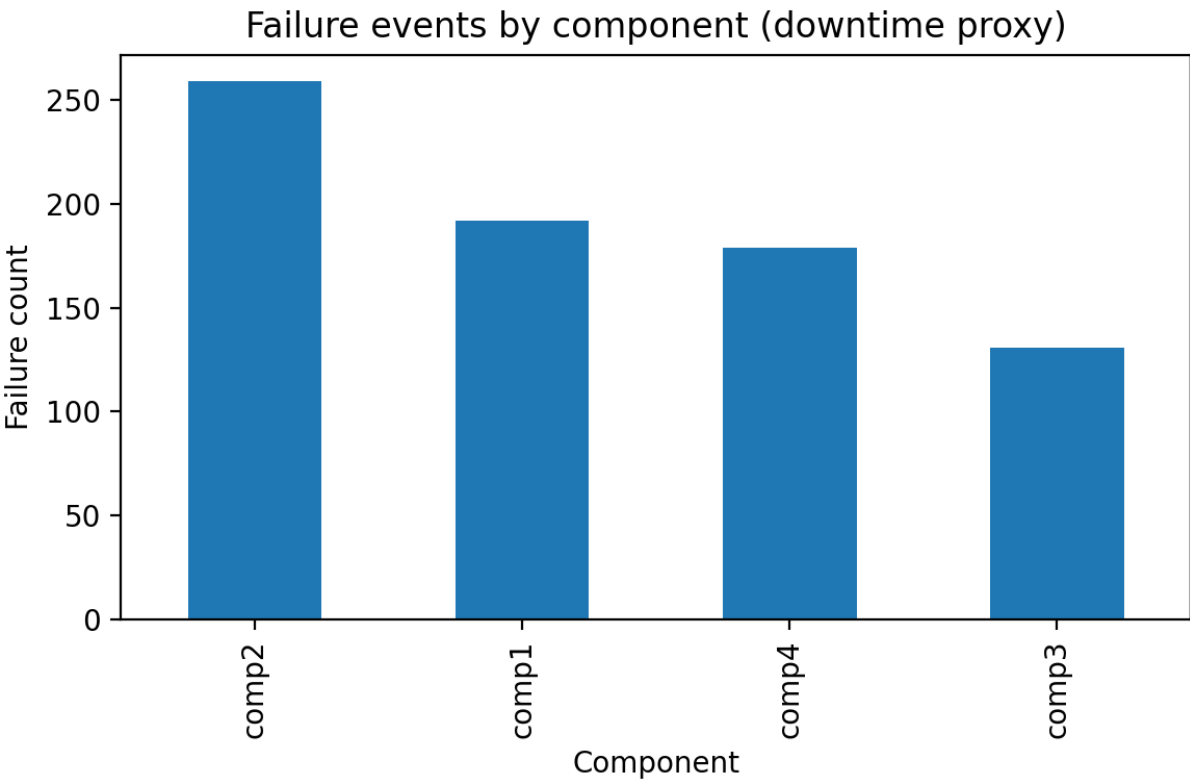
Dataset: Azure Predictive Maintenance (telemetry, errors, maintenance, machines, failures) | Generated: 2026-02-24

Executive summary

This report cleans and analyzes a multi-table predictive maintenance dataset to identify drivers of failure-related downtime and translate patterns into operational recommendations. Since repair duration is not provided, **failure event counts** are used as a **downtime proxy** (more failures typically implies higher expected downtime impact). Key findings: (i) **comp2** accounts for the largest share of failures (34.0%), (ii) **model1** shows the highest failure frequency, (iii) machines **older than 12 years** have ~1.75x higher failure rate, (iv) **vibration increases** strongly in the 24 hours before **comp4** failures, (v) error counts and failure counts are positively correlated ($r=0.54$), and (vi) maintenance timestamps often coincide with failures, suggesting a largely reactive maintenance process.

Q1) Component Downtime Driver (Failure Count Proxy)

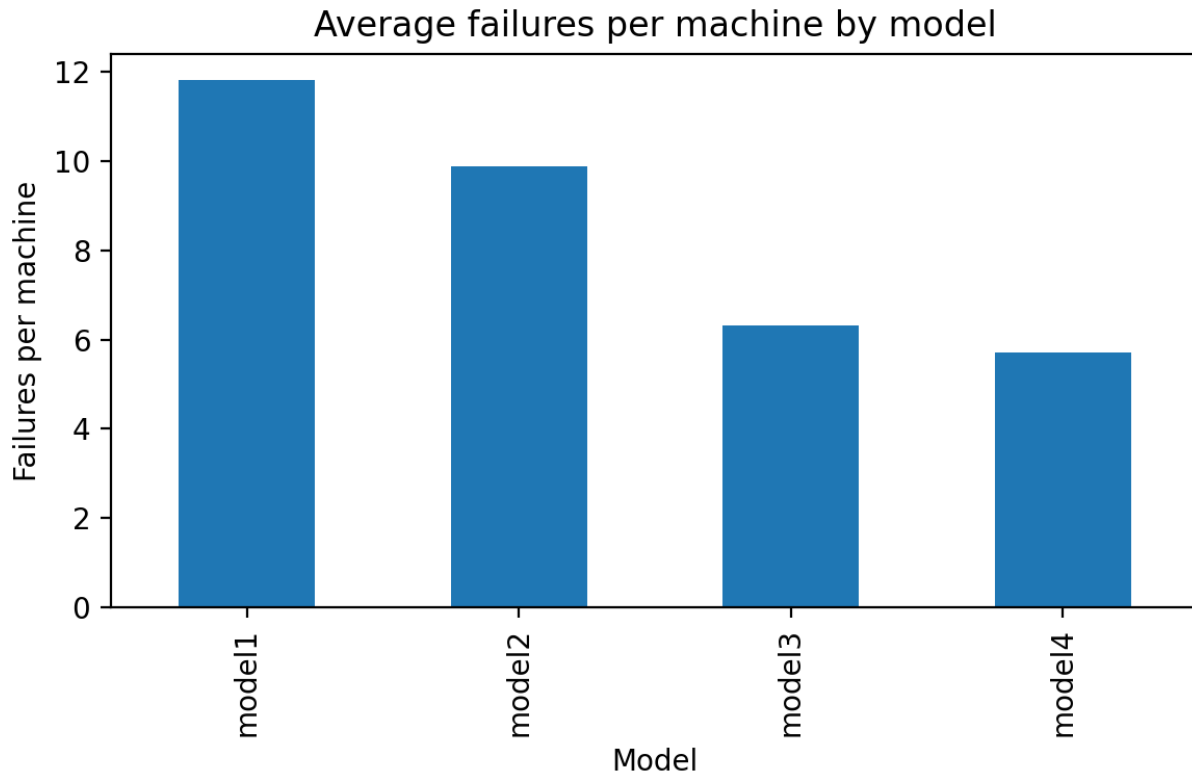
Failure events were aggregated by component to identify the largest driver of downtime risk. **comp2** has the highest count with **259** events (about **34.0%** of all failures, $n=761$). From an operations perspective, reliability programs that reduce comp2 failures (inspection, replacement, supplier quality, or redesign) offer the largest immediate reduction in expected downtime.



Failure events by component. Failure counts are used as a downtime proxy (repair duration not available).

Q2) Model-Level Risk (Failures per Machine)

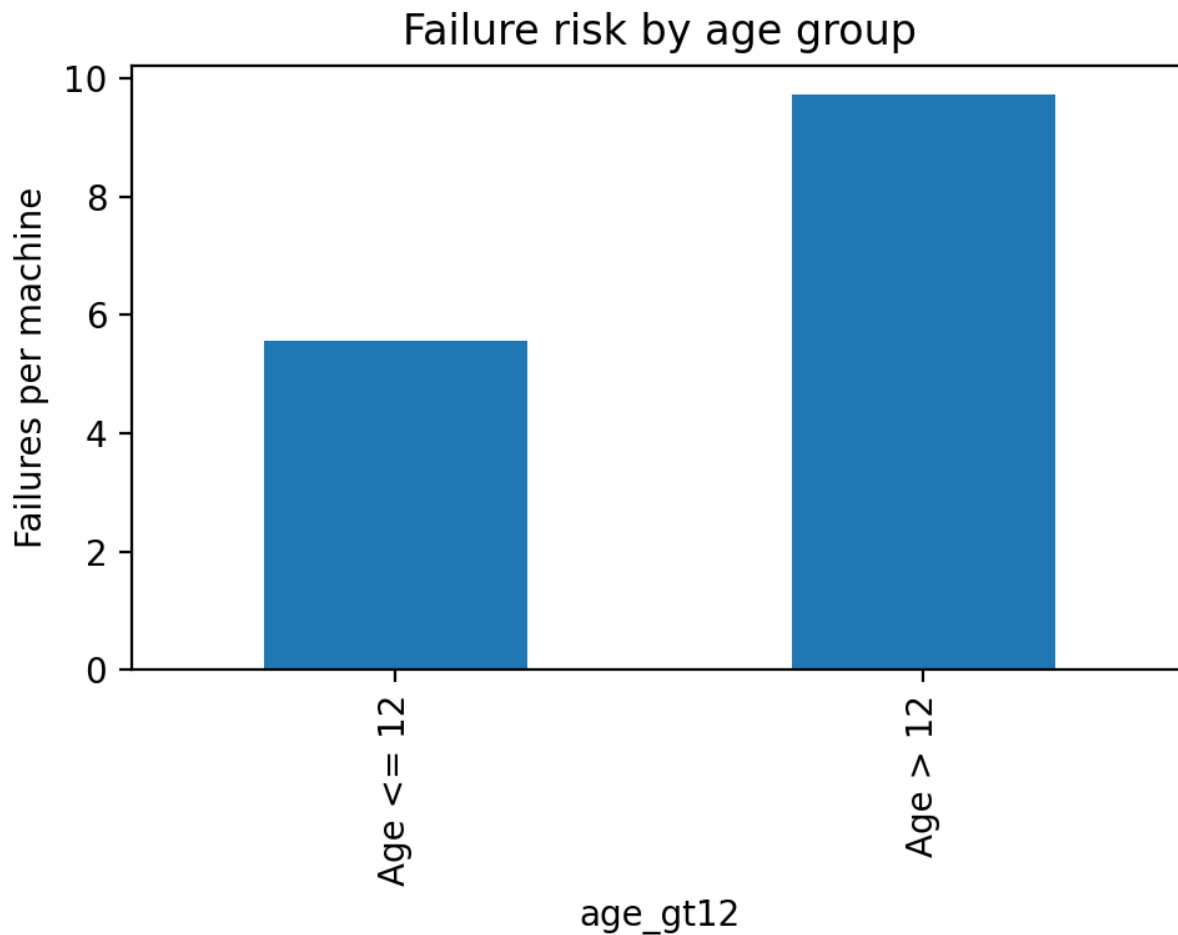
We linked failure events to machine metadata and computed **failures per machine** by model. **model1** has the highest average failure frequency at **11.81** failures per machine, indicating elevated operational risk. This supports model-specific maintenance policies (e.g., tighter inspection intervals for model1).



Average failures per machine by model (normalizes total failures by fleet size per model).

Q3) Age-Related Risk (Older than 12 Years)

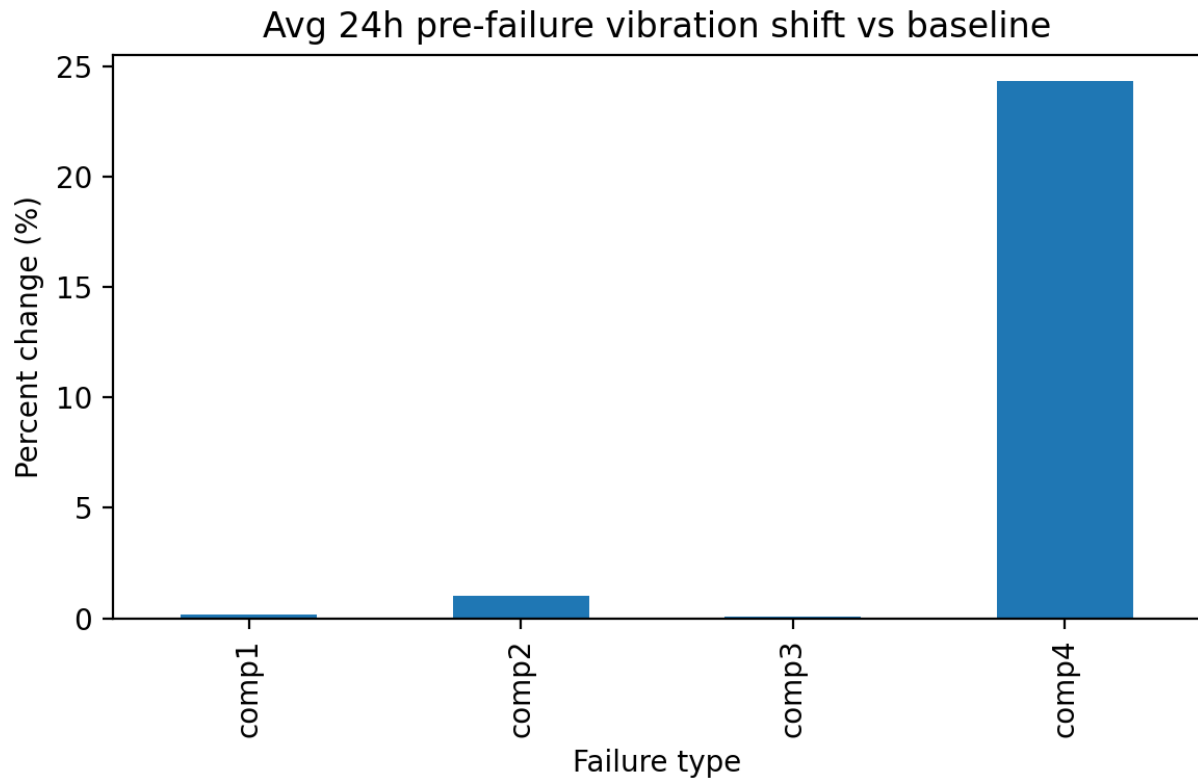
Machines older than 12 years show materially higher failure frequency. The group with **age > 12** averages **9.73** failures per machine versus **5.57** for machines age 12 or younger - a **~1.75x** risk multiplier. Practically, this can inform capital planning (replacement vs. repair) and risk-based scheduling for older assets.



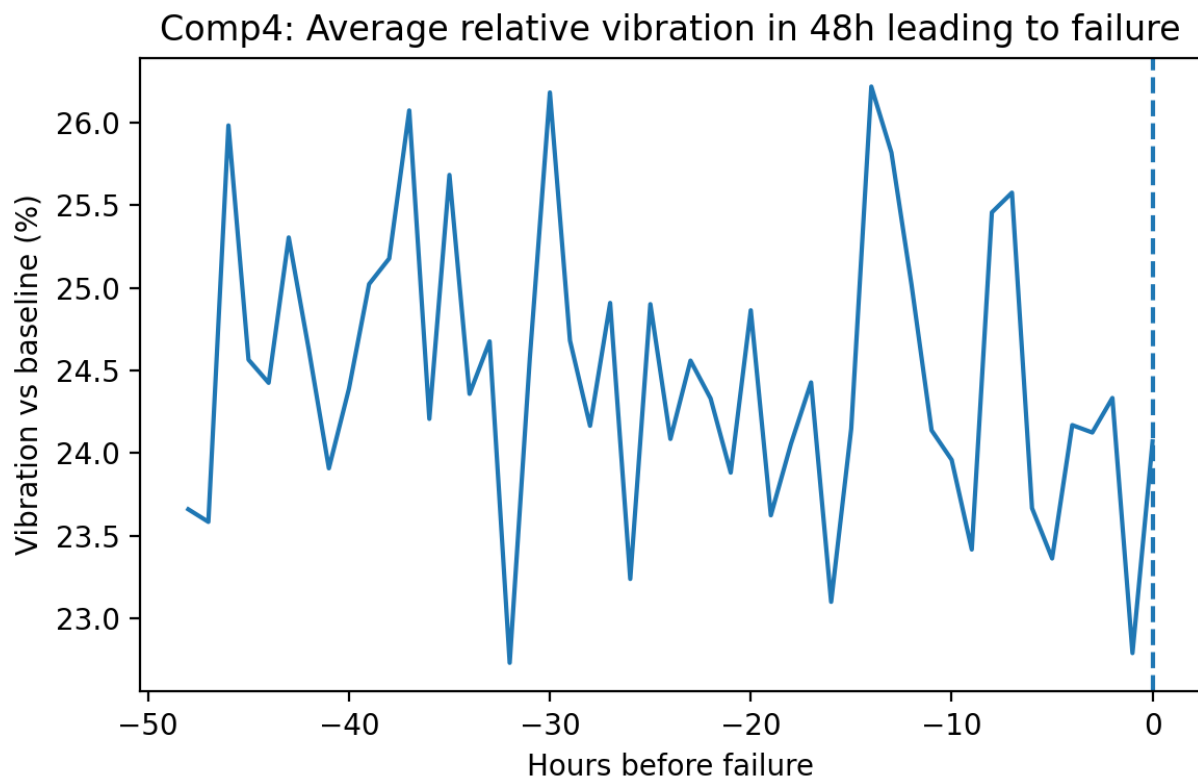
Failure frequency comparison by age group (>12 years vs <=12 years).

Q4) Early-Warning Signal: Vibration Before comp4 Failures

To test for leading indicators, we compared each machine's **24-hour pre-failure** sensor averages with its long-run baseline. For **comp4** failures, vibration rises sharply by about **+24.3%** on average, far larger than other failure types. This pattern supports implementing vibration-based alerts (thresholds or anomaly detection) to trigger inspection before comp4 breakdown.



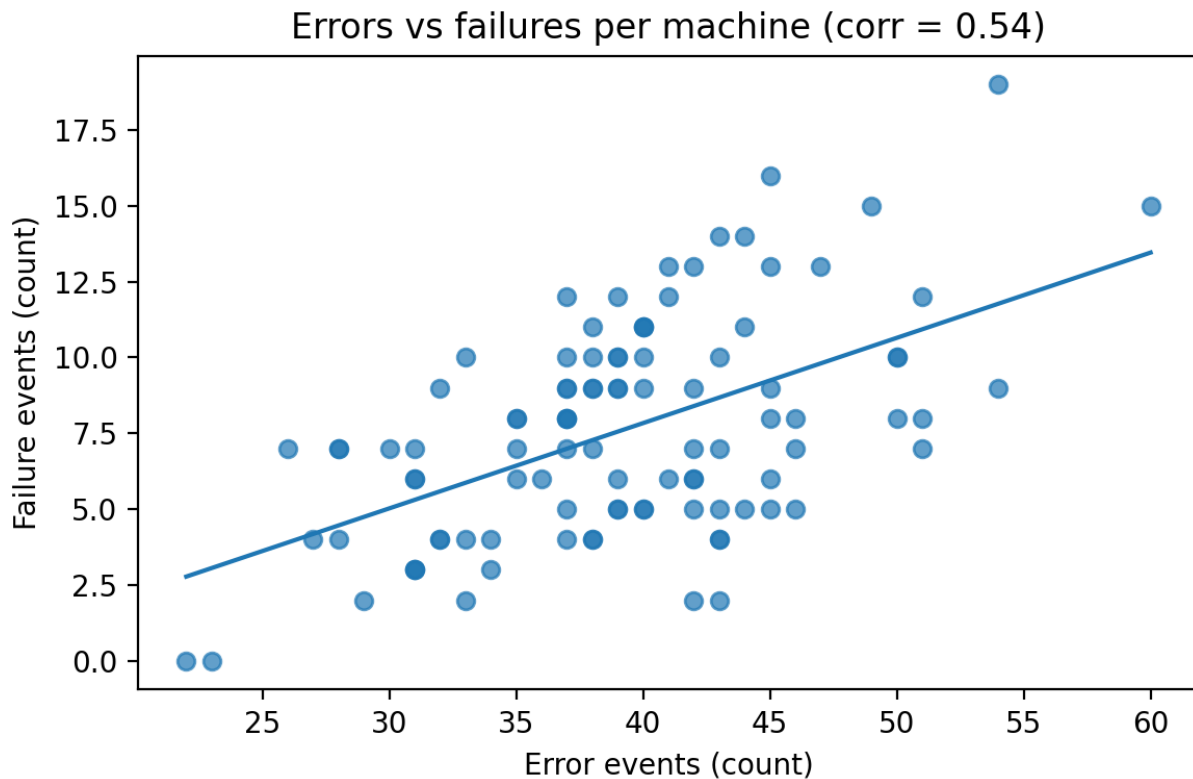
Average 24h pre-failure vibration shift vs machine baseline (percent change).



Comp4 failures: average relative vibration trajectory over the 48 hours leading to failure (normalized by machine baseline).

Q5) Error Logs and Breakdown Probability

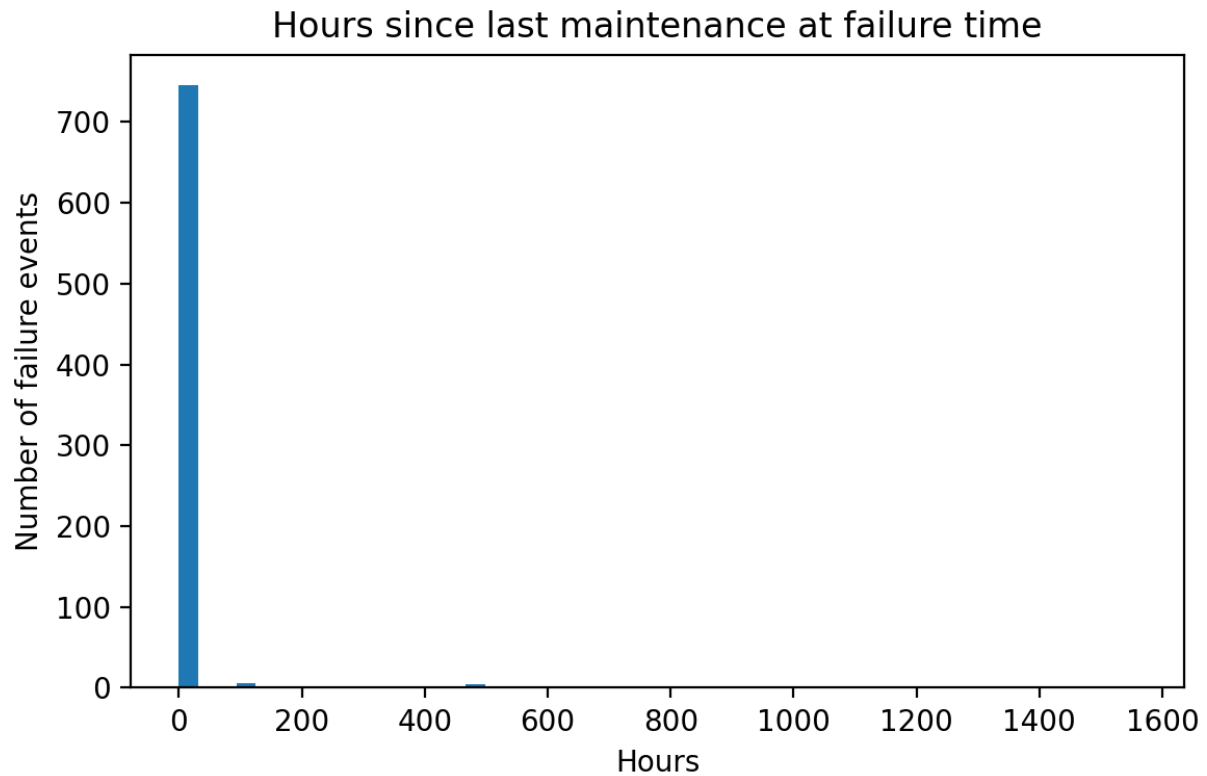
Error events provide operational signals about abnormal behavior. At the **machine level**, total error counts and total failure counts are positively correlated ($r = 0.54$), meaning machines that throw more errors tend to experience more failures. This supports using error volume as a screening feature for identifying high-risk machines for inspection and preventive work orders.



Machine-level relationship between error events and failure events (with linear trend line).

Q6) Maintenance Timing and Process Implication

We estimated time since last maintenance at each failure timestamp. A striking result is that **97.6%** of failures occur at the **same timestamp as a maintenance event**. This indicates maintenance is frequently **corrective/reactive** (performed when the failure occurs) rather than preventive. A practical improvement is to shift maintenance earlier using a risk score driven by: **model (model1)**, **age (>12)**, and **sensor signals** (especially vibration for comp4).



Distribution of hours since last maintenance at the moment of failure. Large mass at 0 hours indicates corrective maintenance.

Operational Recommendations

Priority	Recommendation	Rationale (data-driven)
High	Prioritize comp2 reliability program (inspection/replacement/QC)	comp2 accounts for 34.0% of failure events.
High	Tighten maintenance policy for model1 machines	model1 has the highest failures per machine (11.81).
High	Risk-based scheduling for assets age > 12 years	age>12 has ~1.75x higher failures per machine.
Medium	Deploy vibration-based alerts for comp4	vibration +24.3% in 24h pre-failure window.
Medium	Use error frequency as a screening feature	errors and failures correlate at machine level ($r=0.54$).
Medium	Shift from reactive to preventive maintenance triggers	97.6% failures coincide with maintenance timestamp (reactive pattern).